

Rion Oh
(she/her)

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Department of Physics, KAIST, Daejeon 34141, Republic of Korea

RESEARCH INTERESTS

Galaxy formation and evolution, Baryon cycle, Star formation histories, Quiescent/post-starburst galaxies, Dusty star-forming galaxies

EDUCATION

KAIST (Korea Advanced Institute of Science and Technology)

B.S. in Physics, Minor in Electrical Engineering

Thesis: Diversity of Quenching Pathways in Quiescent Galaxies at $2 \lesssim z \lesssim 6$

Daejeon, South Korea

Expected in Aug 2027

University of Washington

Exchange Student; Research and coursework in the Department of Astronomy

Seattle, WA

Jan 2026 – Aug 2026

PUBLICATIONS

1. **Oh, R.** et al. *LEGGOS: Spatially Resolved Neutral Gas Outflows in SGAS 2111*. 2026, in prep
2. **Oh, R.** et al. *JWST Spectra Conclusively Show an Excess of Neutral Gas Outflows in Quiescent Galaxies at $z=2-5$* . 2026, [arXiv:2609.09459](#)
3. Welch, B., Hutchison, T., [...] **Oh, R.**, et al. *LEGGOS: Direct abundances of N, O, Ne, S, and Ar in five lensed galaxies at Cosmic Noon*. 2026, [arXiv:2609.07851](#)
4. Cleri, N. J., [...] **Oh, R.**, et al. *LEGGOS: A Shocking Lack of Evidence for Shocks at sub-kiloparsec Scales at $2 < z < 4$* . 2026, [arXiv:2609.07850](#)
5. Khullar, G., [...] **Oh, R.**, et al. *LEGGOS I: The JWST LEGGOS Survey – LEnsing and Galaxy Growth: Observing Substructures – Unpacks the Nature of Clumpy Star Formation and Quenching in Gravitationally Lensed Galaxies beyond Cosmic Noon*. 2026, [arXiv:2606.20845](#)

RESEARCH EXPERIENCE

Neutral Gas Outflows in High-Redshift Quenching Galaxies

Dec 2025–present

Red Galaxies Group, University of Washington, Advisor: Prof. Arianna S.Long

- Performed Prospector SED fitting and spectral stacking of 274 galaxies at $z=2-5$ to Na I D absorption
- Detected neutral gas outflows in all quiescent bins, with outflow properties 0.5–2 dex above non-quiescent
- Showed outflow energetics exceed stellar feedback, implicating AGN-driven quenching

Spatially Resolved Neutral Gas Outflows in Lensed Galaxies

Apr 2026–present

JWST LEGGOS Team, Advisor: Dr. Gourav Khullar

- Extracting spatially resolved, clump-level spectra from JWST observations of SGAS2111
- Measuring Na I D absorption to constrain neutral gas flows across individual clumps
- Connecting clump-level gas kinematics to physical properties and evolutionary phase

Diversity of Quenching Pathways

Dec 2025–present

Observational Cosmology & Astrophysics Laboratory, KAIST, Advisor: Prof. Junhan Kim

- Reconstructed SFHs of $2 \leq z \leq 6$ quiescent galaxies using JWST PRISM spectra and photometry
- Classified galaxies into pathways, revealing trends in mass, metallicity, dust, and burst fraction

- Interpreted subgroup trends as evidence for multiple pathways and heterogeneous baryon histories

Triaxial Modeling of Intracluster Gas with X-ray and SZ Observations Jun–Dec 2025

Observational Cosmology & Astrophysics Laboratory, KAIST, *Advisor: Prof. Junhan Kim*

- Modeled intracluster gas using joint X-ray and SZ observations within the CLUMP-3D framework
- Implemented triaxial density and pressure models projected along the line of sight
- Performed Bayesian MCMC inference on a 13-parameter cluster model

AWARDS & HONORS

Dean’s List, *University of Washington* Winter, Spring 2026

Korea–U.S. Advanced Fields Youth Scholar, *KIAT & IIE*, national selection (200 awardees) 2026

Astra Recon Award (satellite), Hanwha Aerospace & Systems Space Challenger, top 5 teams 2025

Outstanding Student for Academic Inquiry, *KAIST Physics*, 2 students (astrophysics) Fall 2025

2nd Place, Scientific Writing Award (English), *KAIST* 2024

Academic Excellence Scholarship, *KAIST*, merit-based 2024–2027

Institutional Merit Scholarship, *KAIST*, full-tuition 2023–2027

Science & Technology Talent Scholarship, *KAIST*, merit-based 2023

PRESENTATION

Extragalactic Astronomy Symposium, *University of Washington* Apr 2026

Oral: “Statistical Detection of Neutral Gas Outflows in $z = 2$ –5 Quenching Galaxies with JWST”

Undergraduate Research Symposium, *University of Washington* Apr 2026

Poster: “Statistical Detection of Neutral Gas Outflows in $z = 2$ –5 Quenching Galaxies with JWST”

RESEARCH TRAINING & PROJECTS

Extragalactic Astronomy & Cosmology Summer School Jul 2025

Korea Institute for Advanced Study

- SIDM cosmological simulations with *MUSIC & GIZMO*: core fraction analysis as a function of halo mass
- Galaxy merger simulations with *DICE & GIZMO*: dark matter, gas, and stellar dynamics
- Radio telescope receiver circuit design; antenna parameter constraints for 21 cm intensity mapping

Radio Summer School & Radio Telescope Users’ Meeting Aug 2025

Korea Astronomy and Space Science Institute

- Constructed and operated a small radio telescope; performed data acquisition and analysis
- Visited KVN data center; attended Users Meeting covering major facilities (ALMA, EAVN, VLBI)

Stratospheric Balloon Payload Project, *KAIST* Feb–Nov 2025

- Built a radiosonde payload with custom circuitry and sensor integration
- Modeled trajectories to predict landing coordinates across two launches

OBSERVATION

Apache Point Observatory 3.5m Telescope, KOSMOS Observing Run (5 hours) May 14, 2026

- Spectroscopic follow-up of lensing candidates using g-band emission lines for redshift constraints

NYSC 1m Telescope, Deokheung Optical Astronomy Observatory (6 hours) Apr 01, 2025

- Time-series photometry of Kepler-17b transits; extracted light curves and derived orbital parameters

TEACHING EXPERIENCES

Instructor, Elementary STEM Outreach Program Feb–Jun 2025

Instructor, Secondary-Level Physics Mar–Dec 2024

Tutor, Undergraduate STEM Subjects (General Physics, Calculus, General Chemistry) Feb 2023–Dec 2025

SERVICES & OUTREACH

Founder & Author, *The Young Universe* Newsletter Apr 2026–present

Mentor, Program for Deaf Youth Feb 2024–present

Student Council Member, Department of Physics, KAIST Feb 2024–Jan 2026

Department Representative, Department of Physics, KAIST Feb 2024–Jan 2025

REFERENCES

Prof. Arianna S. Long, Department of Astronomy, University of Washington (aslong@uw.edu)

Dr. Gourav Khullar, Department of Astronomy, University of Washington (gkhullar@uw.edu)

Prof. Junhan Kim, Department of Physics, KAIST (junhan@kaist.ac.kr)